

EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 10
Sinusoidal Steady-State
Analysis

Sinusoidal Steady-State Analysis

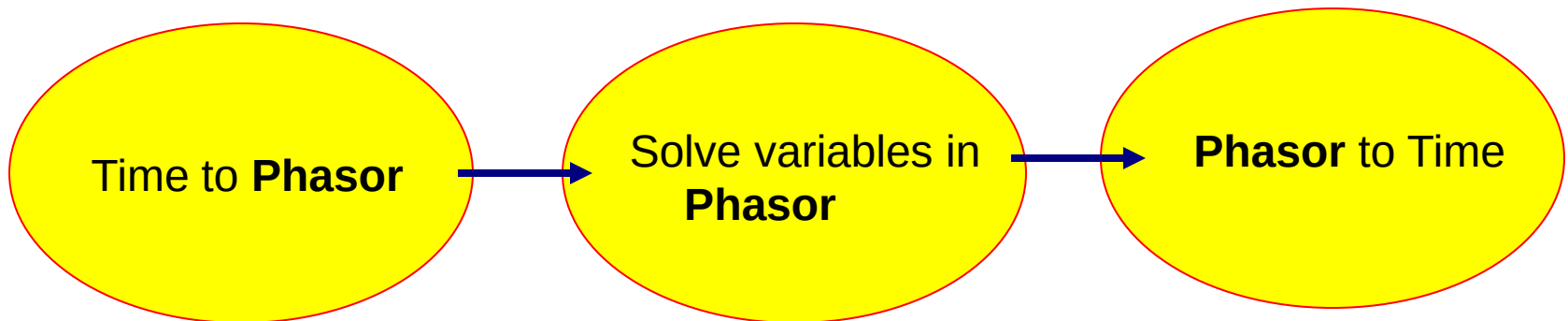
Chapter 10

- 10.1 Basic Approach
- 10.2 Nodal Analysis
- 10.3 Mesh Analysis
- 10.4 Superposition Theorem
- 10.5 Source Transformation
- 10.6 Thevenin and Norton Equivalent Circuits

10.1 Basic Approach (1)

Steps to Analyze AC Circuits:

1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.



10.2 Nodal Analysis

Example 10.1

Determine i_x .

$$jX_L = j\omega L = j \times 4 \times 1 = j4$$

$$jX_L = j\omega L = j \times 4 \times 0.5 = j2$$

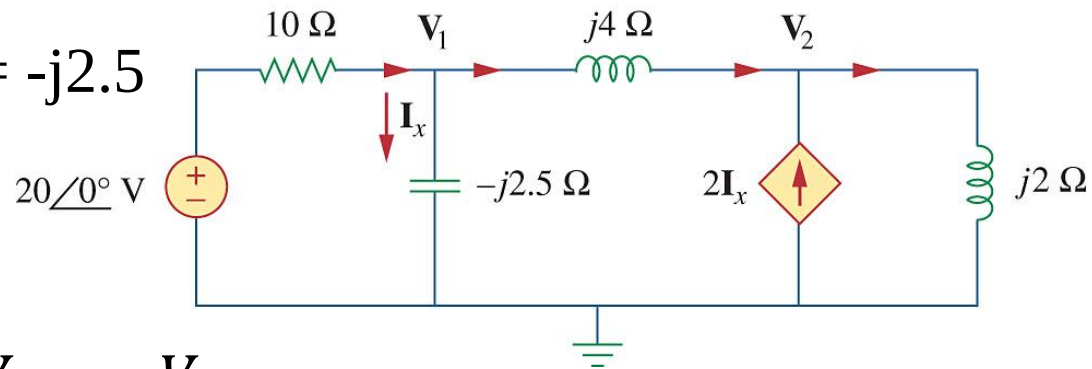
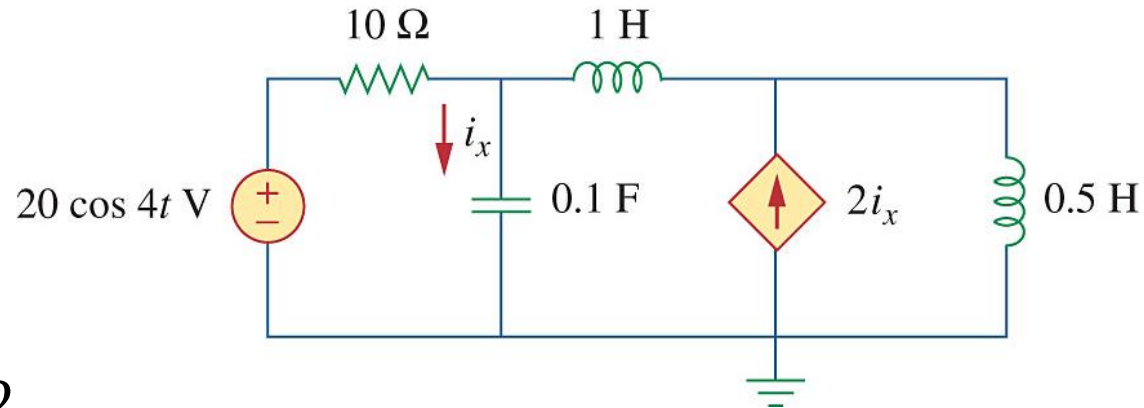
$$jX_C = -j/\omega C = -j \div (4 \times 0.1) = -j2.5$$

KCL at node 1:

$$\frac{V_1 - 20\angle 0^\circ}{10} + \frac{V_1 - V_2}{j4} + \frac{V_1}{-j2.5} = 0$$

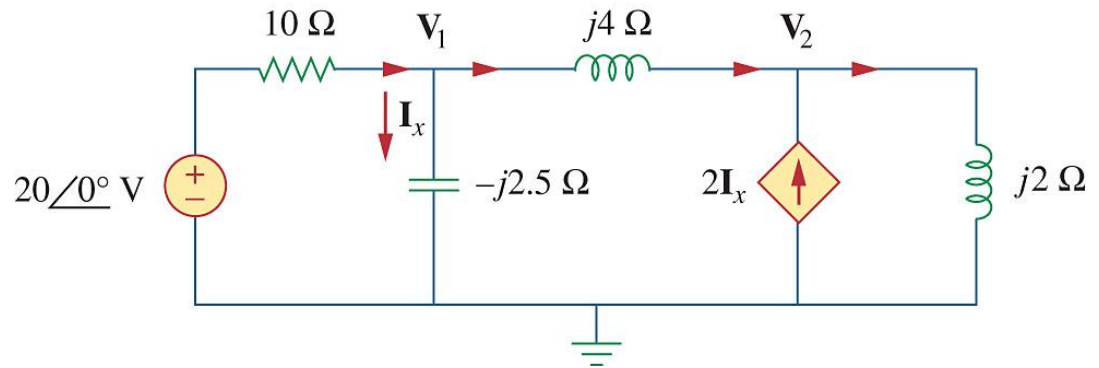
$$(\times 20) \quad 2V_1 - 40 - j5(V_1 - V_2) + j8V_1 = 0$$

$$(2 + j3)V_1 + j5V_2 = 40 \quad \textcircled{1}$$



Example 10.1 cont

Determine i_x .



KCL at node 2:

$$\frac{V_2 - V_1}{j4} - 2I_x + \frac{V_2}{j2} = 0$$

But: $I_x = \frac{V_1}{-j2.5}$ hence: $\frac{V_2 - V_1}{j4} - 2\left(\frac{V_1}{-j2.5}\right) + \frac{V_2}{j2} = 0$

$$(\times j20) \quad 5V_2 - 5V_1 + 16V_1 + 10V_2 = 0$$

$$11V_1 + 15V_2 = 0 \quad (2)$$

Example 10.1 cont

Determine i_x .

$$\begin{bmatrix} 2 + j3 & j5 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \end{bmatrix}$$

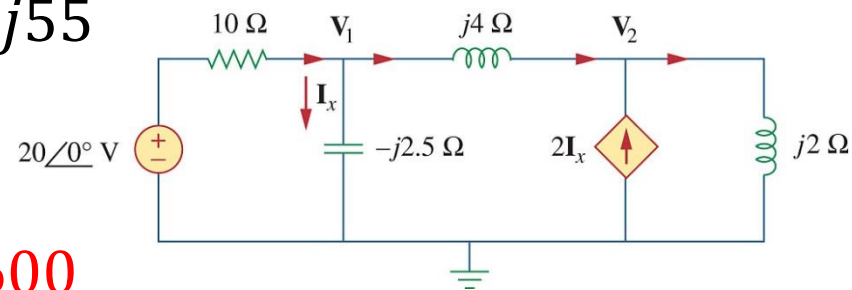
$$\begin{aligned} \Delta &= \begin{vmatrix} 2 + j3 & j5 \\ 11 & 15 \end{vmatrix} = (2 + j3) \cdot 15 - j5 \cdot 11 \\ &= (30 + j45) - j55 \\ &= 30 - j10 \end{aligned}$$

$$\Delta_1 = \begin{vmatrix} 40 & j5 \\ 0 & 15 \end{vmatrix} = 40 \cdot 15 - 0 = 600$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{600}{31.62 \angle -18.43^\circ} = 18.97 \angle 18.43^\circ \text{ V}$$

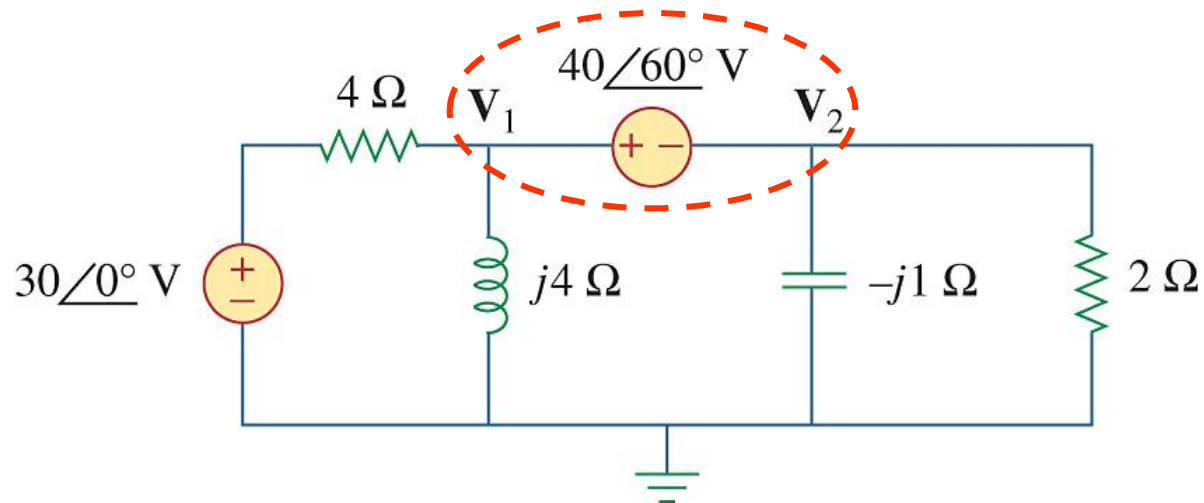
$$I_x = \frac{V_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.43^\circ \text{ A}$$

$$i_x = \underline{7.59 \cos(4t + 108.43^\circ) \text{ A}}$$



PP 10.2 (Super Node)

Determine V_1 and V_2 .



KCL at Super node:

$$\frac{V_1 - 30\angle 0^\circ}{4} + \frac{V_1}{j4} + \frac{V_2}{-j1} + \frac{V_2}{2} = 0$$

$$(\times 4) \quad V_1 - 30 - jV_1 + j4V_2 + 2V_2 = 0$$

$$(1 - j)V_1 + (2 + j4)V_2 = 30 \quad (1)$$

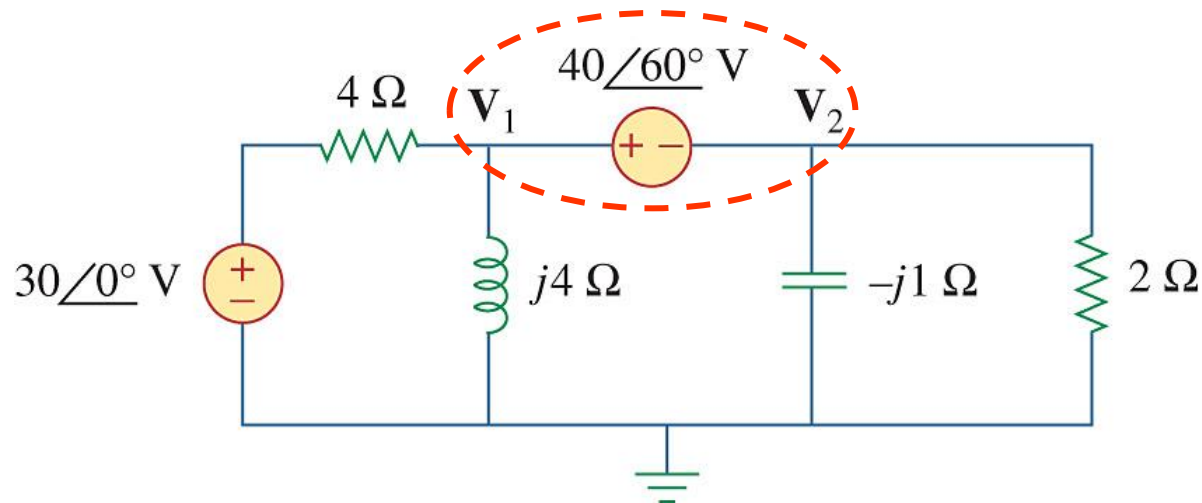
Super node: $V_1 - V_2 = 40\angle 60^\circ$

$$V_1 = V_2 + 40\angle 60^\circ \quad (2)$$

PP 10.2 cont ...

Determine V_1 and V_2 .

② in ①



$$(1 - j)(V_2 + 40\angle 60^\circ) + (2 + j4)V_2 = 30$$

$$(1 - j + 2 + j4)V_2 + (1 - j)(40\angle 60^\circ) = 30$$

$$(3 + j3)V_2 + (\sqrt{2}\angle -45^\circ)(40\angle 60^\circ) = 30$$

$$(\sqrt{18}\angle 45^\circ)V_2 = 30 - (\sqrt{2} \cdot 40\angle 15^\circ)$$

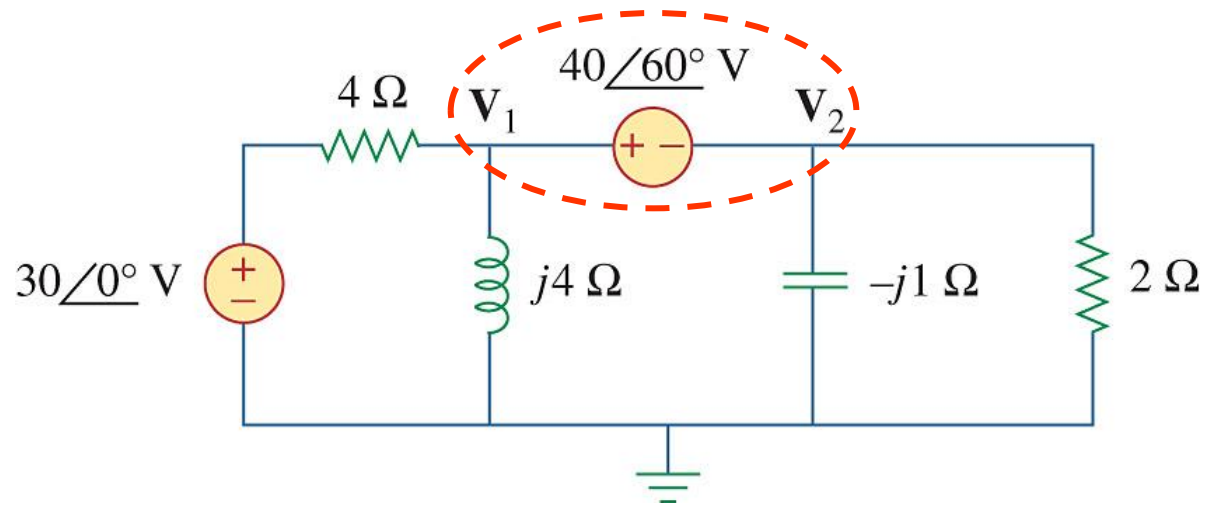
$$V_2 = \frac{30 - 54.64 - j14.64}{\sqrt{18}\angle 45^\circ} = \frac{28.66\angle -149.28^\circ}{4.24\angle 45^\circ}$$

$$\underline{V_2 = 6.76\angle 165.72^\circ \text{ V}} \quad \text{③}$$

PP 10.2 cont ...

Determine V_1 and V_2 .

③ in ②



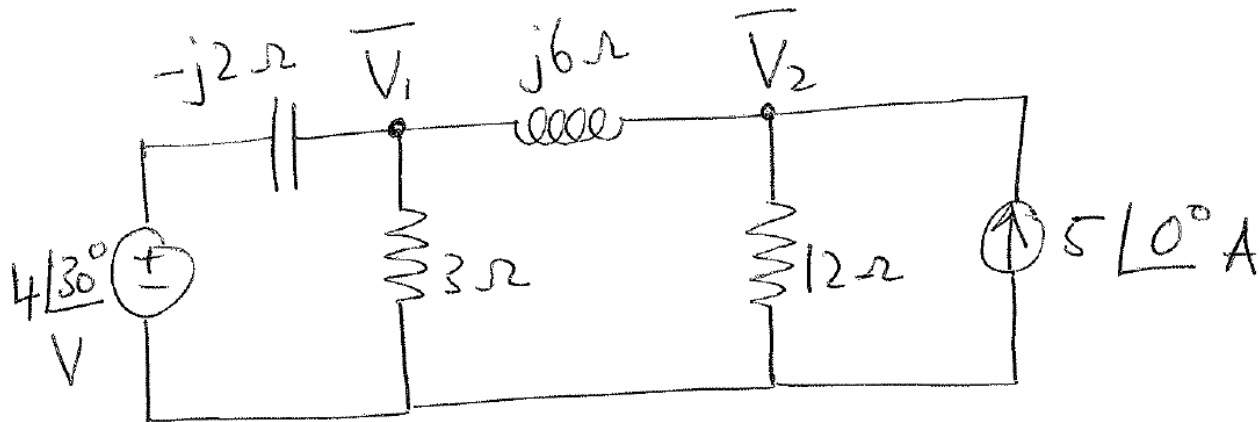
$$\begin{aligned} V_1 &= V_2 + 40\angle 60^\circ \\ &= 6.76\angle 165.72^\circ + 40\angle 60^\circ \\ &= -6.547 + j1.6 + 20 + j34.641 \\ &= 13.45 + j36.31 \\ &= \underline{38.72\angle 69.67^\circ \text{ V}} \end{aligned}$$

Example 1:

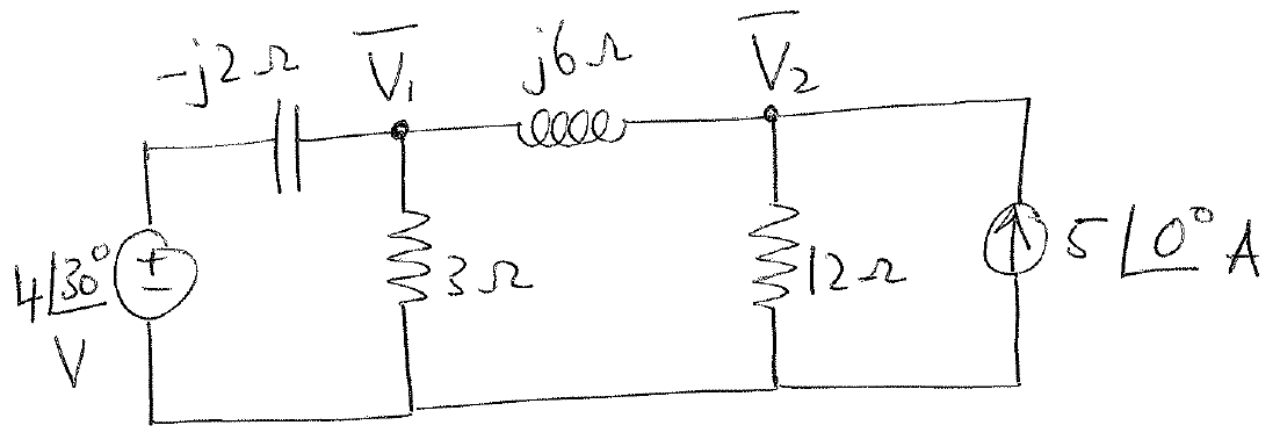
Set up the nodal analysis equations for $\mathbf{V}_1$ and $\mathbf{V}_2$. Write it in the following form:

$$(a_1 + jb_1)\mathbf{V}_1 + (a_2 + jb_2)\mathbf{V}_2 = A_3 \angle \phi_3$$

$$(a_4 + jb_4)\mathbf{V}_1 + (a_5 + jb_5)\mathbf{V}_2 = A_6 \angle \phi_6$$



Exercise:



KCL at Node 1:

$$\frac{V_1 - 4\angle 30^\circ}{-j2} + \frac{V_1}{3} + \frac{V_1 - V_2}{j6} = 0$$

$$(\times 6) \quad j3 \cdot (V_1 - 4\angle 30^\circ) + 2V_1 + (-j)(V_1 - V_2) = 0$$

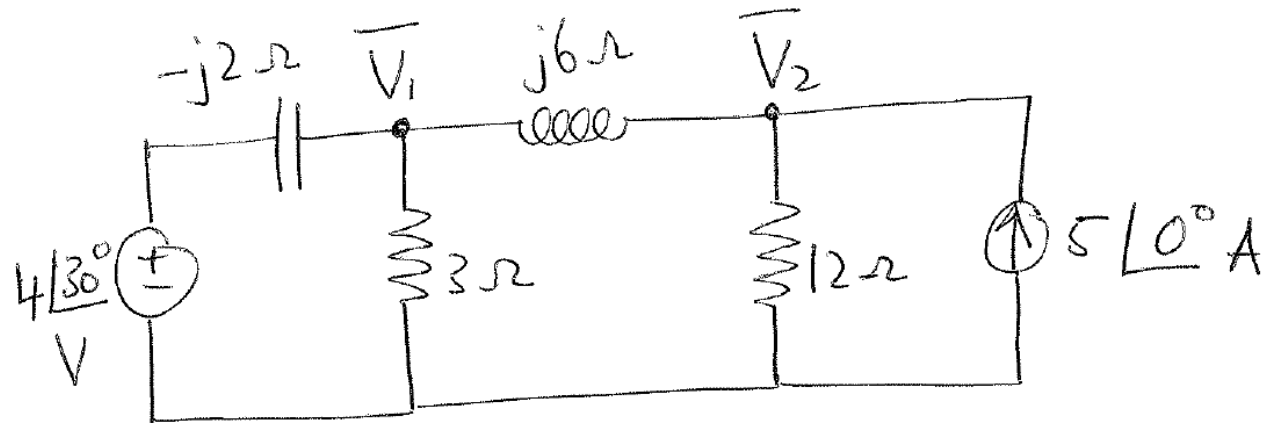
$$j3V_1 + 2V_1 - jV_1 + jV_2 = 12\angle 120^\circ$$

$$(2 + j2)V_1 + jV_2 = 12\angle 120^\circ$$

$$(a_1 + jb_1)V_1 + (a_2 + jb_2)V_2 = A_3\angle\phi_3$$

$a_1 = 2; \quad b_1 = 2; \quad a_2 = 0; \quad b_2 = 1; \quad A_3 = 12; \quad \phi_3 = 120^\circ$
--

Exercise cont...



KCL at Node 2:

$$\frac{V_2 - V_1}{j6} + \frac{V_2}{12} - 5\angle 0^\circ = 0$$

$$(\times 12) \quad (-j2)(V_2 - V_1) + V_2 - 60\angle 0^\circ = 0$$

$$-j2V_2 + j2V_1 + V_2 = 60\angle 0^\circ$$

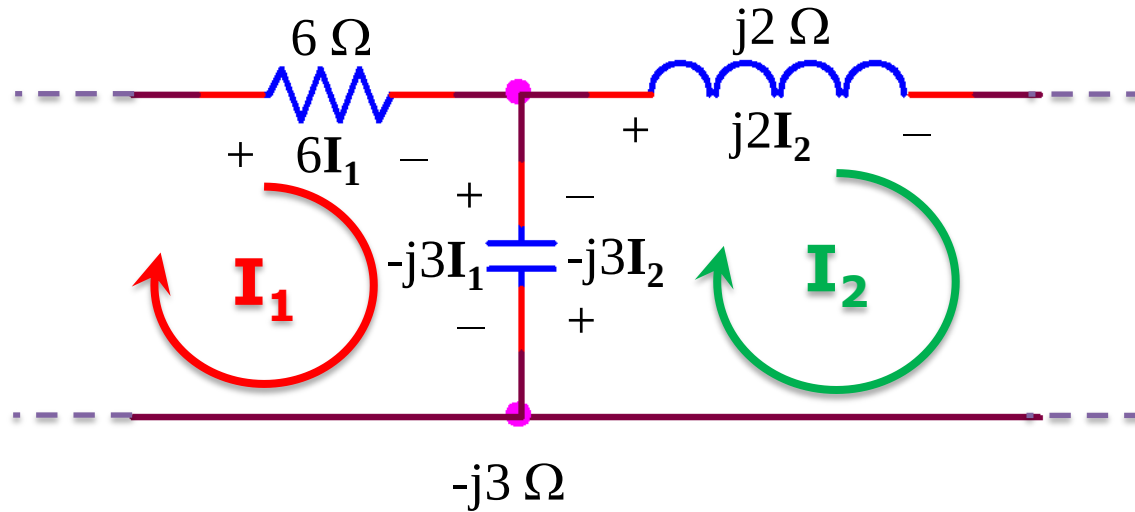
$$j2V_1 + (1 - 2j)V_2 = 60\angle 0^\circ$$

$$(a_4 + jb_4)V_1 + (a_5 + jb_5)V_2 = A_6\angle\phi_6$$

$a_4 = 0; \quad b_4 = 2; \quad a_5 = 1; \quad b_5 = -2; \quad A_6 = 60; \quad \phi_6 = 0^\circ$

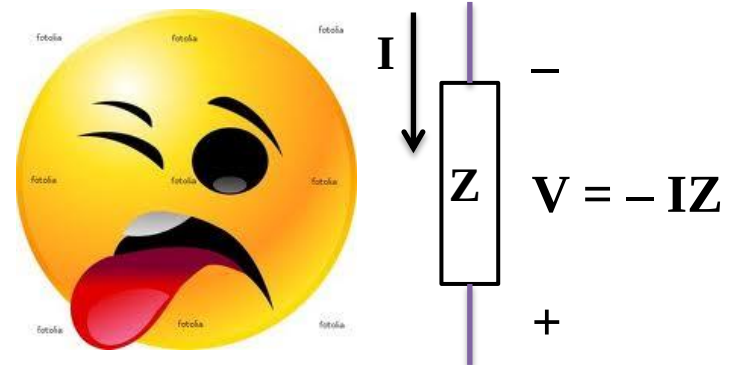
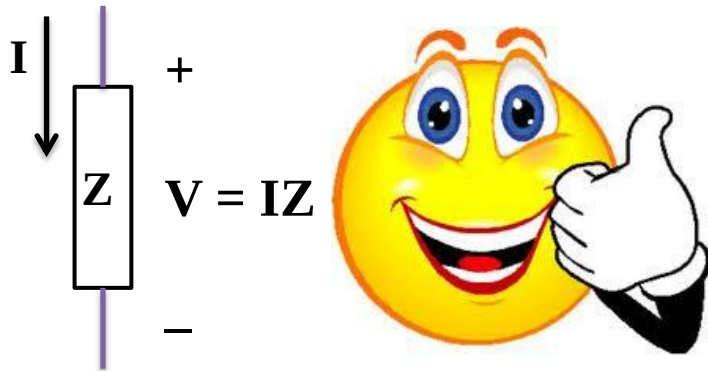
Ch. 10 L#2

10.3 Mesh Analysis



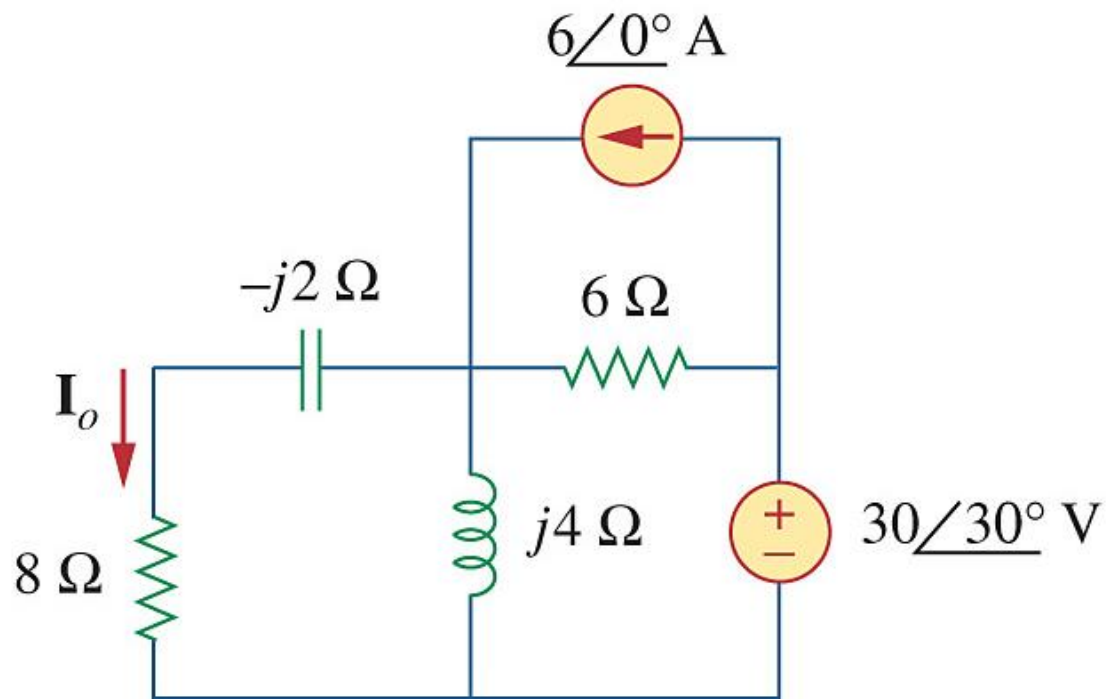
$$\dots + 6I_1 + (-j3)(I_1 - I_2) = 0$$

$$\dots + j2I_2 + (-j3)(I_2 - I_1) = 0$$



PP 10.3

Find $\mathbf{I_o}$.



PP 10.3

Find $\mathbf{I_o}$.

KVL Loop 1:

$$8I_1 + (-j2)I_1 + j4(I_1 - I_2) = 0$$

$$(8 + j2)I_1 - j4I_2 = 0$$

$$I_2 = \frac{(8 + j2)I_1}{j4}$$

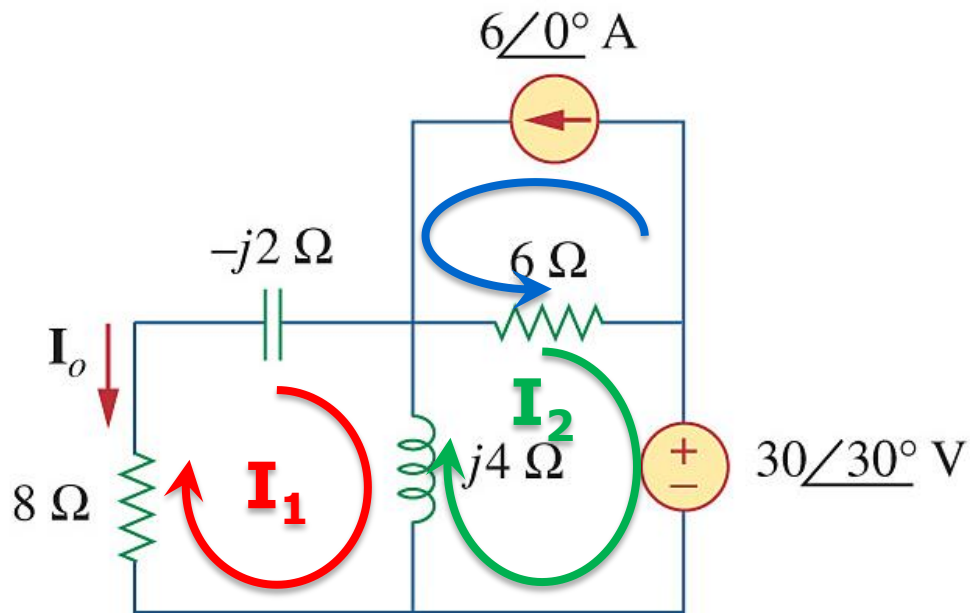
$$= (-j2 + 0.5)I_1 \quad (1)$$

KVL Loop 2:

$$j4(I_2 - I_1) + 6(I_2 + 6\angle 0^\circ) + 30\angle 30^\circ = 0$$

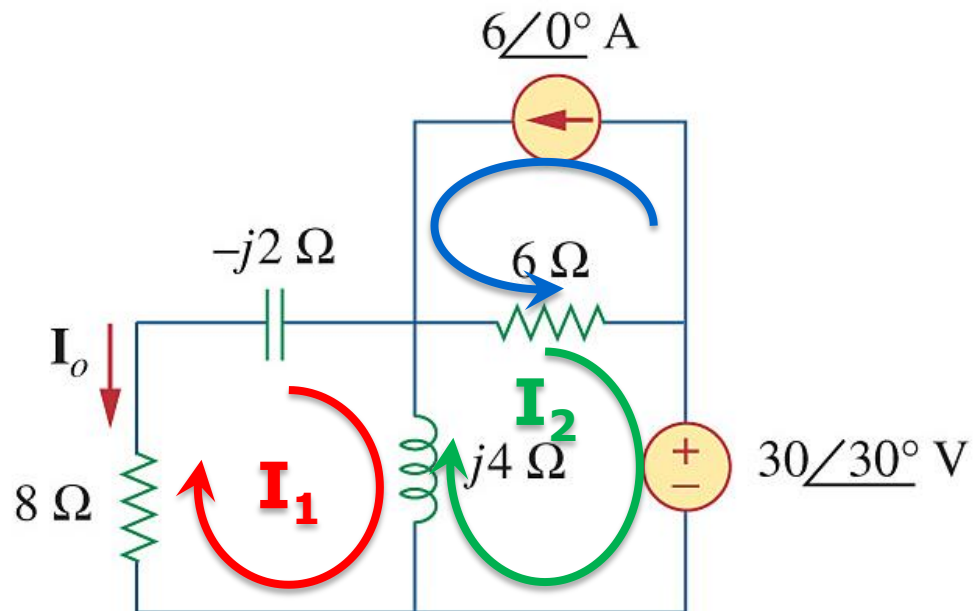
$$(-j4)I_1 + (6 + j4)I_2 = -36\angle 0^\circ - 30\angle 30^\circ$$

$$(-j4)I_1 + (6 + j4)I_2 = -62 - j15 = 63.77 \angle -166.4^\circ \quad (2)$$



PP 10.3 cont...

Find I_o .



① in ② :

$$(-j4)I_1 + (6 + j4)(-j2 + 0.5)I_1 = 63.77 \angle -166.4^\circ$$

$$(-j4)I_1 + (-12j + 3 + 8 + j2)I_1 = 63.77 \angle -166.4^\circ$$

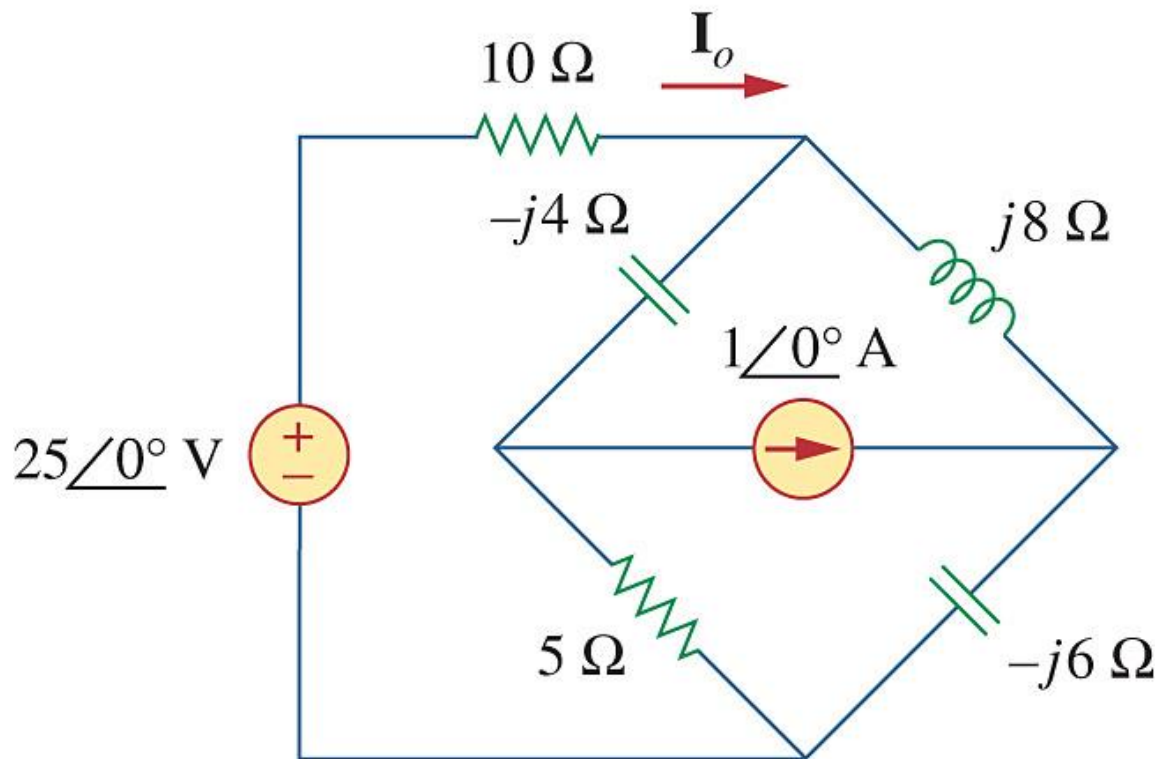
$$(11 - j12)I_1 = 63.77 \angle -166.4^\circ$$

$$I_1 = \frac{63.77 \angle -166.4^\circ}{17 \angle -51.84^\circ} = 3.58 \angle -114.55^\circ$$

$$I_o = -I_1 = \underline{3.58 \angle 65.45^\circ \text{ A}}$$

PP 10.4

Find $\mathbf{I_o}$.



PP 10.4

Find I_o .

Super Mesh: $I_2 - I_1 = 1\angle 0^\circ$

$$I_2 = 1\angle 0^\circ + I_1 \quad (1)$$

Loop O:

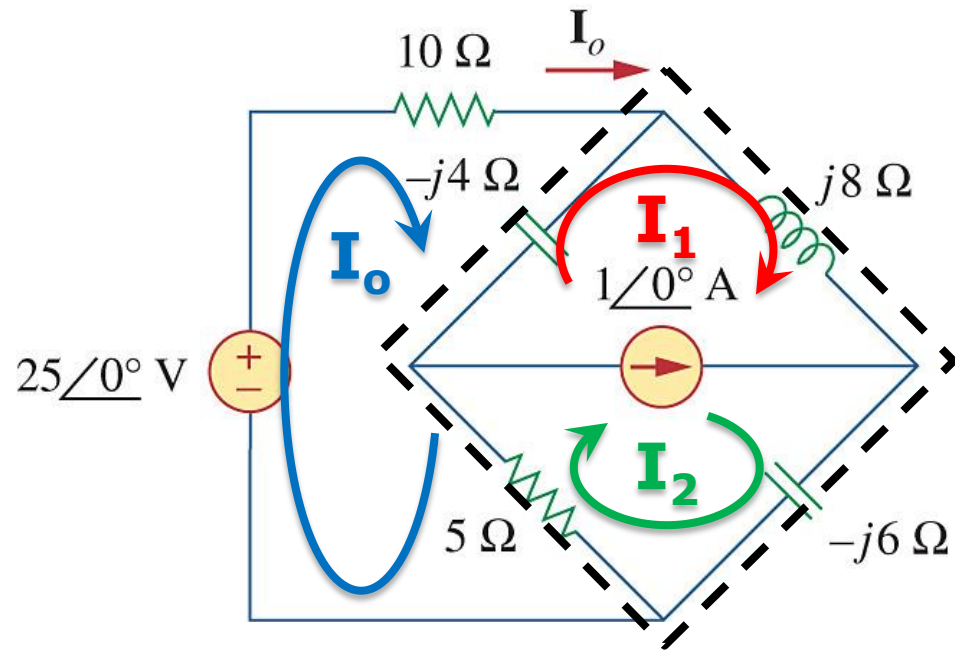
$$-25\angle 0^\circ + 10I_o + (-j4)(I_o - I_1) + 5(I_o - I_2) = 0$$

$$(15 - j4)I_o + j4I_1 - 5I_2 = 25\angle 0^\circ$$

$$(15 - j4)I_o + j4I_1 - 5(1 + I_1) = 25\angle 0^\circ$$

$$(15 - j4)I_o + (-5 + j4)I_1 = 30\angle 0^\circ$$

$$I_1 = \frac{30\angle 0^\circ - (15 - j4)I_o}{(-5 + j4)} \quad (2)$$



PP 10.4 cont...

Find I_o .

Super Mesh: $I_2 - I_1 = 1\angle 0^\circ$

$$I_2 = 1\angle 0^\circ + I_1 \quad (1)$$

Super Mesh KVL:

$$5(I_2 - I_o) + (-j4)(I_1 - I_o) + j8I_1 - j6I_2 = 0$$

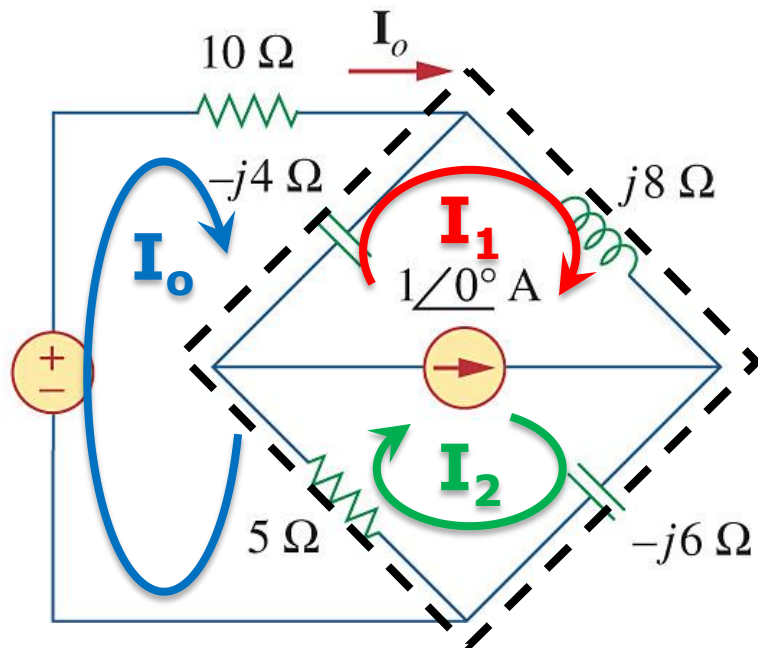
$$(-5 + j4)I_o + j4I_1 + (5 - j6)I_2 = 0$$

$$(-5 + j4)I_o + j4I_1 + (5 - j6)(1 + I_1) = 0$$

$$(-5 + j4)I_o + (5 - j2)I_1 = (-5 + j6) \quad (3)$$

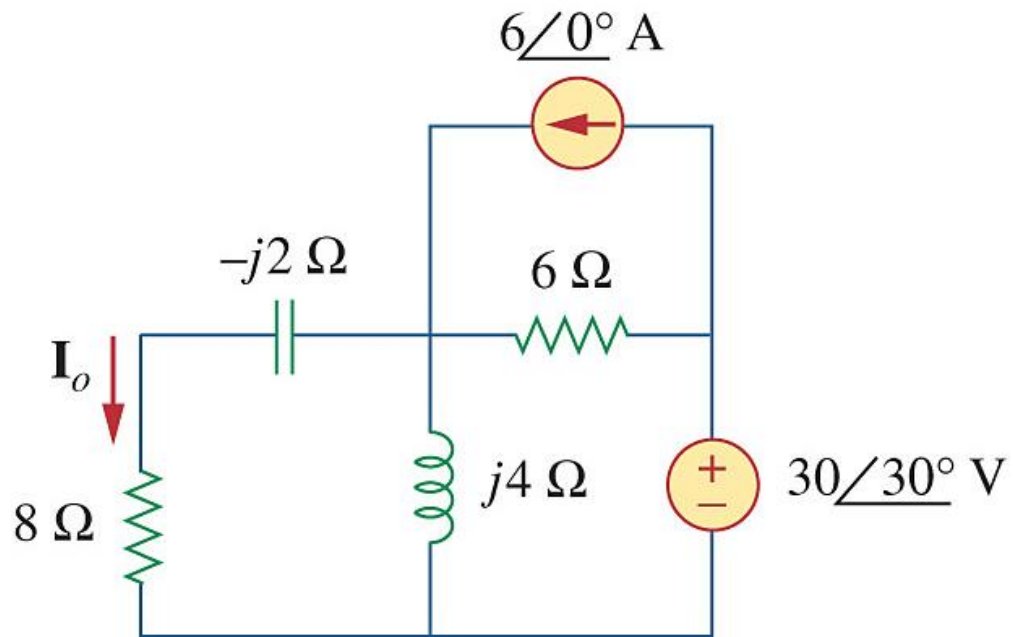
$$(2) \text{ in } (3) : (-5 + j4)I_o + (5 - j2) \left(\frac{30\angle 0^\circ - (15 - j4)I_o}{(-5 + j4)} \right) = (-5 + j6)$$

$$I_o = \underline{2.54\angle 5.94^\circ \text{ A}}$$



PP 10.5 (Super Positioning)

Find I_o .



PP 10.5 (Super Positioning)

Find I_o .

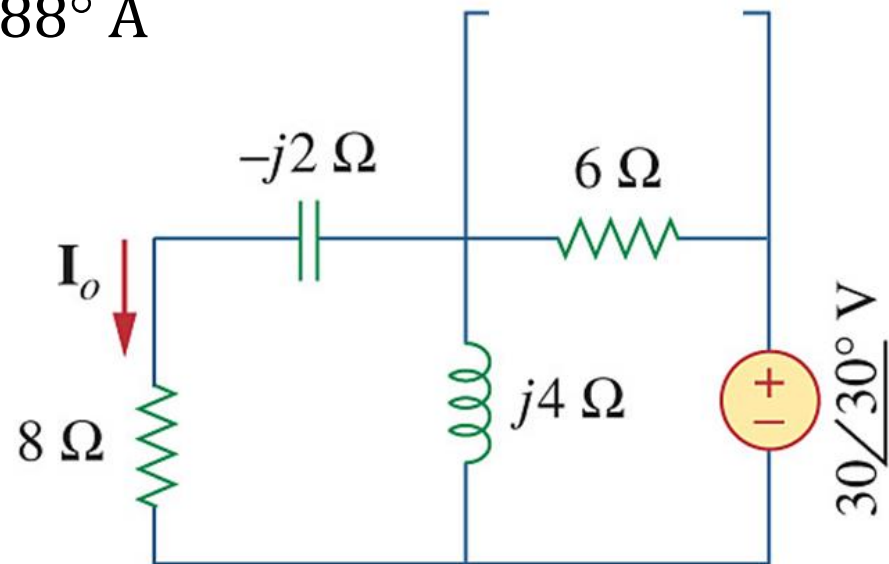
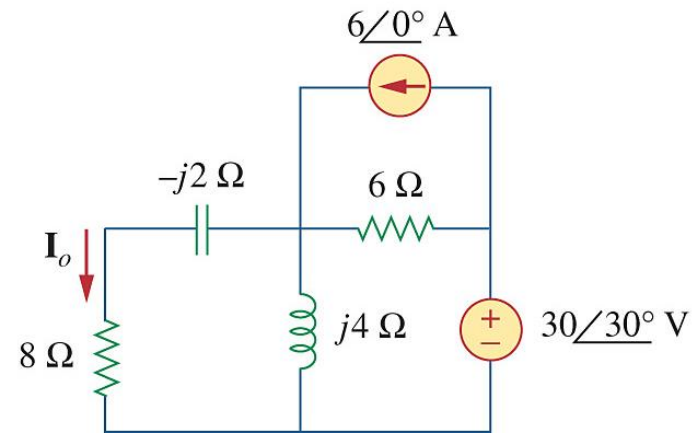
I_o due to $30\angle 30^\circ$ V source:

$$Z_{\text{tot}} = (8 - j2) // j4 + 6 = 8.636 \angle 24.121^\circ \Omega$$

$$I_{\text{tot}} = \frac{30\angle 30^\circ}{8.636\angle 24.121^\circ} = 3.474\angle 5.88^\circ \text{ A}$$

$$I_{o1} = 3.474\angle 5.88^\circ \frac{j4}{8 - j2 + j4}$$

$$I_{o1} = \underline{1.685\angle 81.843^\circ \text{ A}}$$



PP 10.5 cont

Find I_o .

I_o due to $6\angle 0^\circ$ A source:

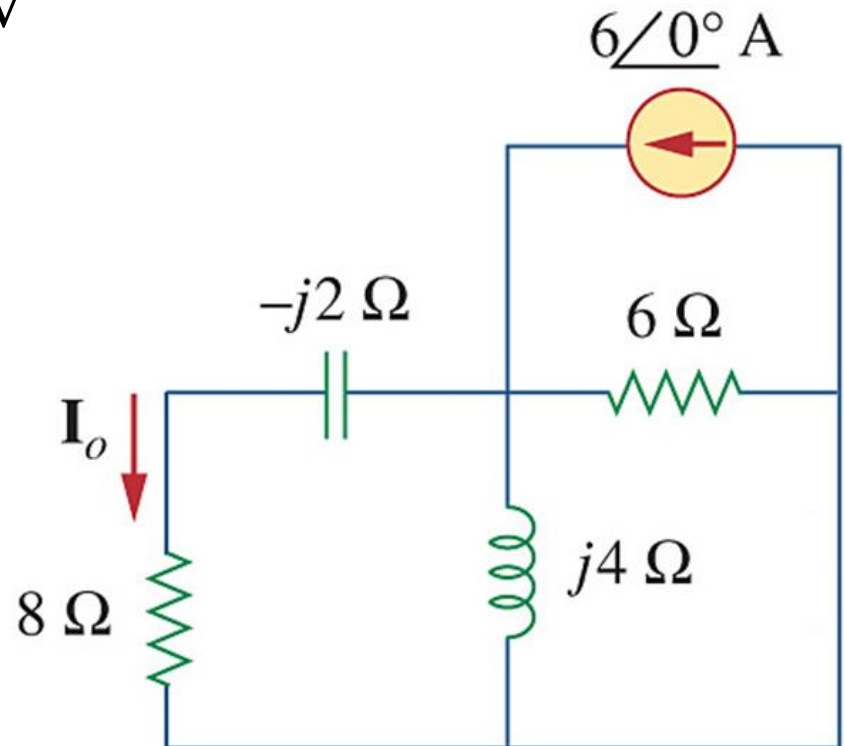
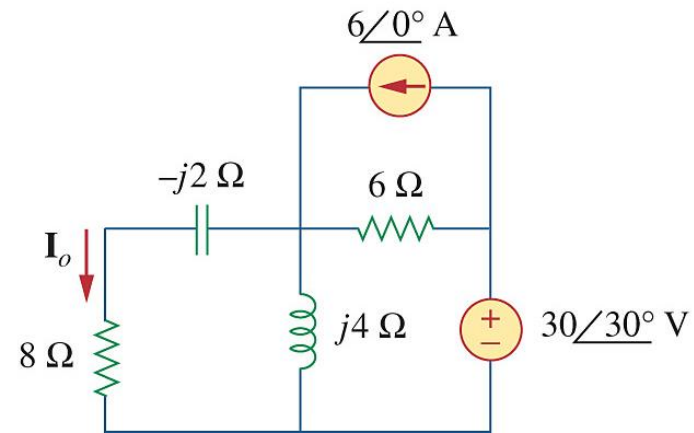
$$Z_{\text{tot}} = (8 - j2) \parallel j4 \parallel 6 = 2.779 \angle 37.806^\circ \Omega$$

$$V_{\parallel} = 6\angle 0^\circ \times Z_{\text{tot}} = 16.674 \angle 37.806^\circ \text{ V}$$

$$I_{o2} = \frac{16.674 \angle 37.806^\circ}{8 - j2}$$

$$I_{o2} = \underline{2.022 \angle 51.843^\circ \text{ A}}$$

$$I_o = I_{o1} + I_{o2} = \underline{3.58 \angle 65.45^\circ \text{ A}}$$

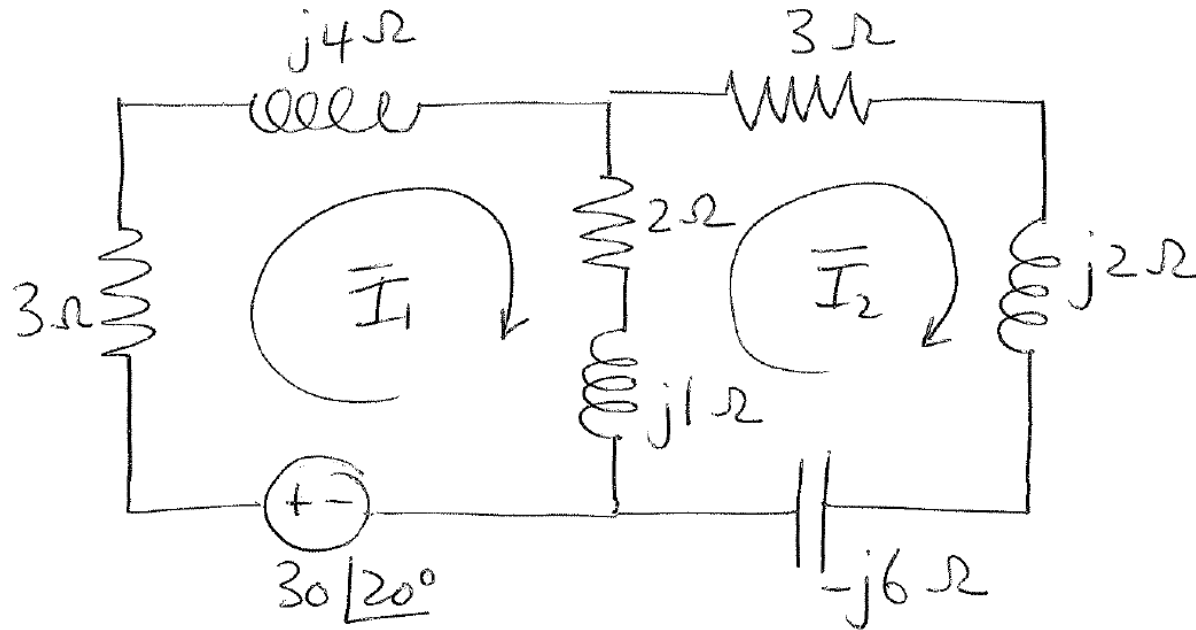


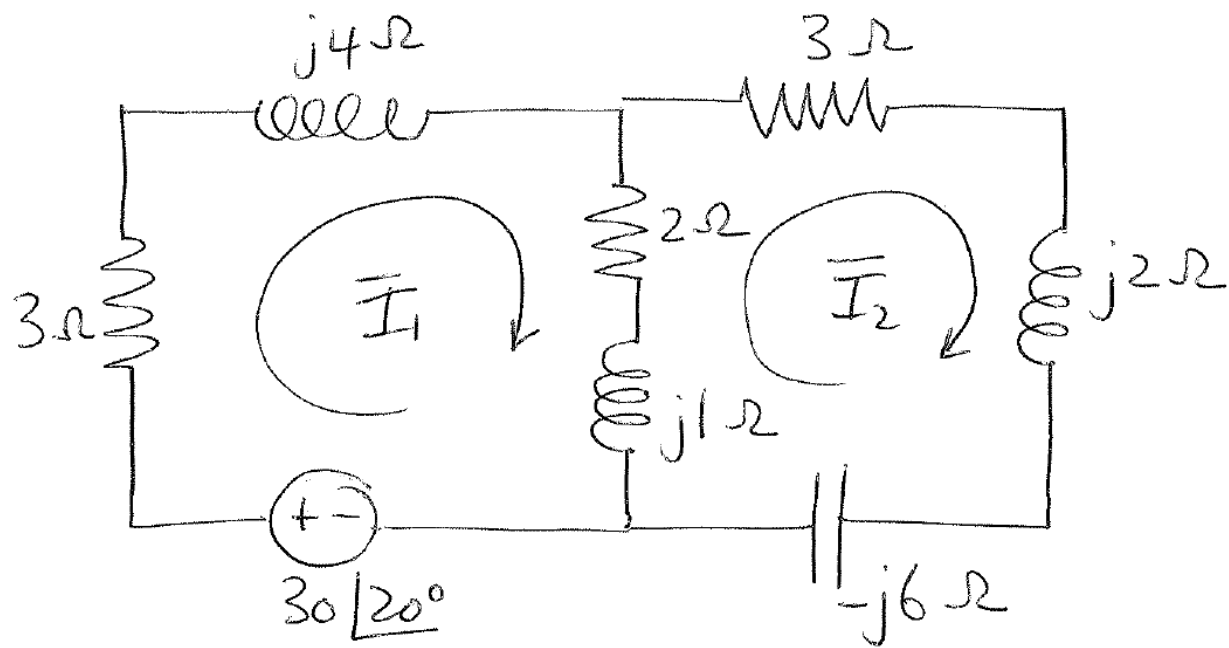
Example 2:

Set up the mesh equations for $\mathbf{I}_1$ and $\mathbf{I}_2$. Write it in the following form:

$$(a_1 + jb_1)\mathbf{I}_1 + (a_2 + jb_2)\mathbf{I}_2 = A_3 \angle \phi_3$$

$$(a_4 + jb_4)\mathbf{I}_1 + (a_5 + jb_5)\mathbf{I}_2 = A_6 \angle \phi_6$$

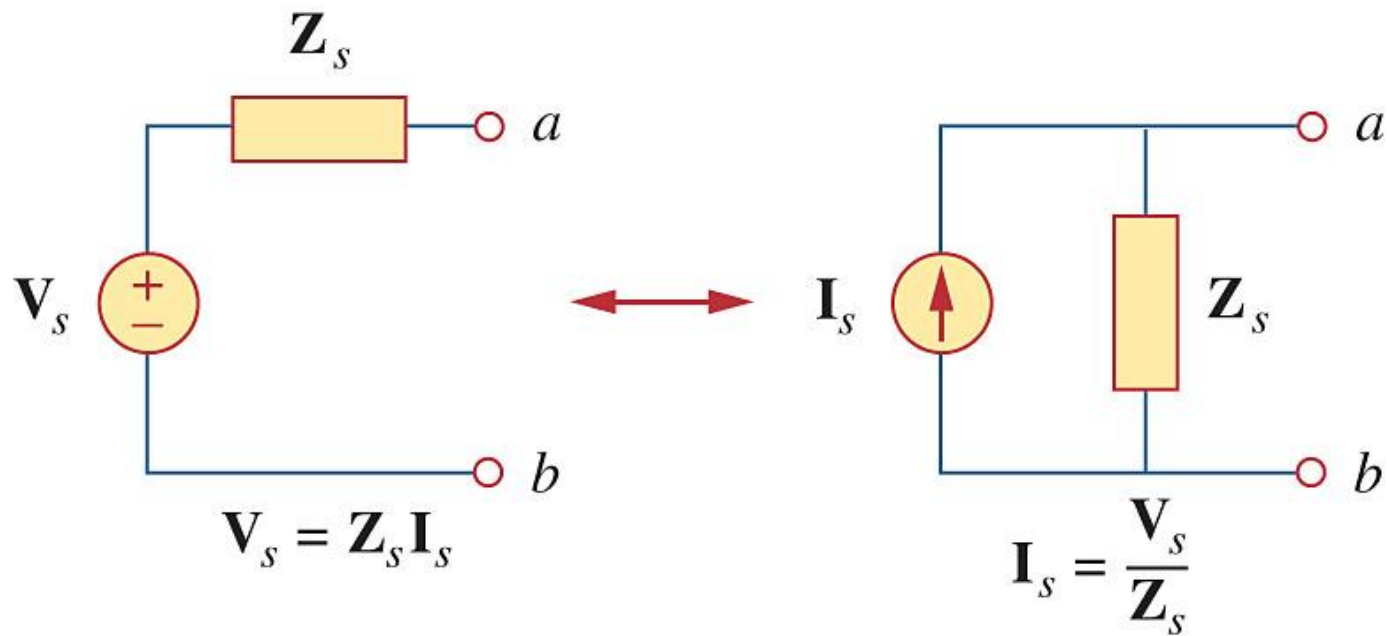




Ch. 10 L#3

10.5 Source Transformation

10.6 Thevenin & Norton

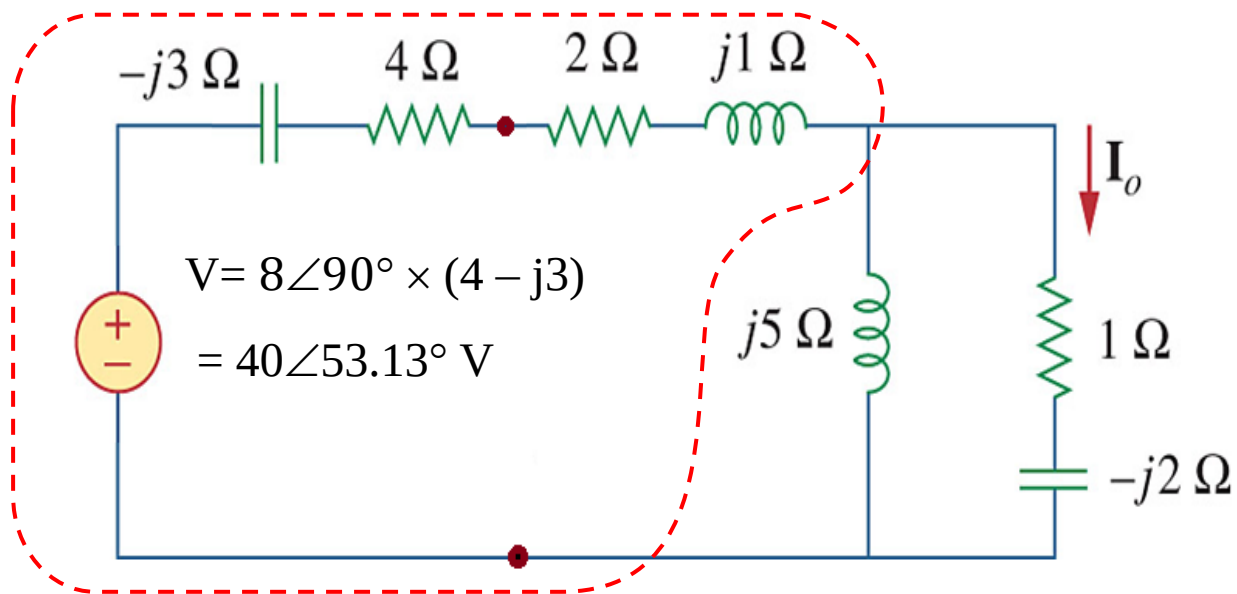
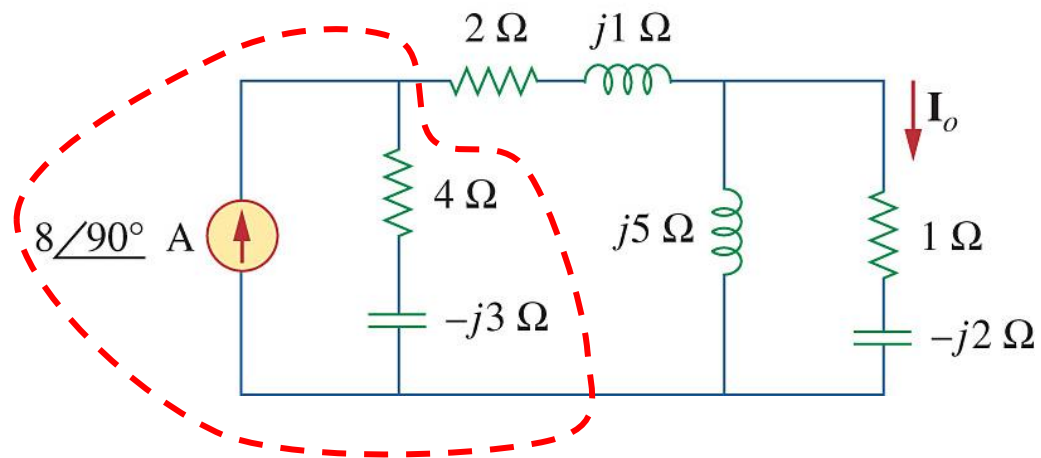


Thevenin Equivalent

Norton Equivalent

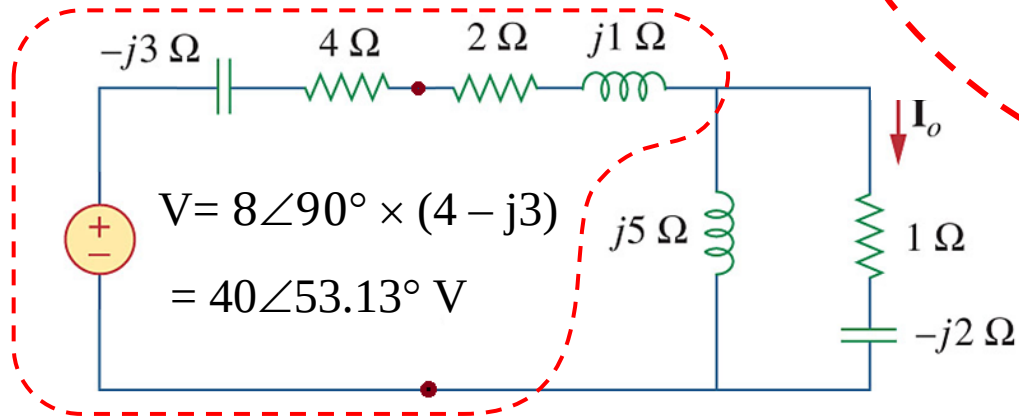
PP 10.7

Find $\mathbf{I}_o$ using source transformation

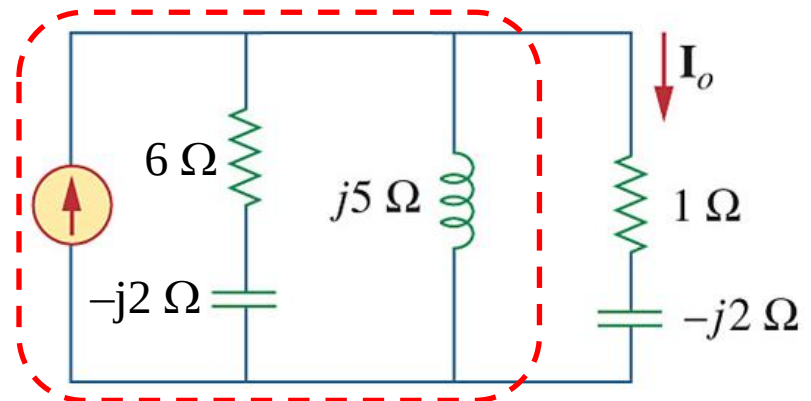
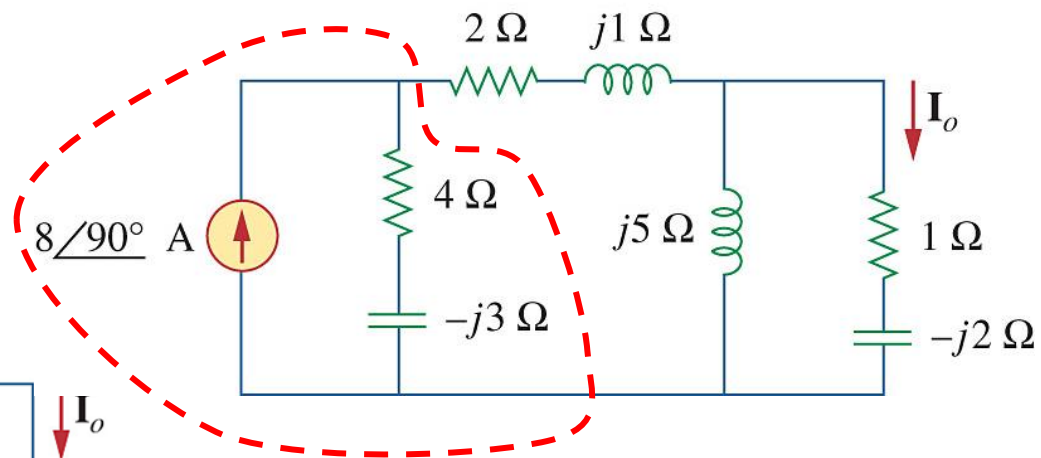


PP 10.7

Find $\mathbf{I}_o$ using source transformation

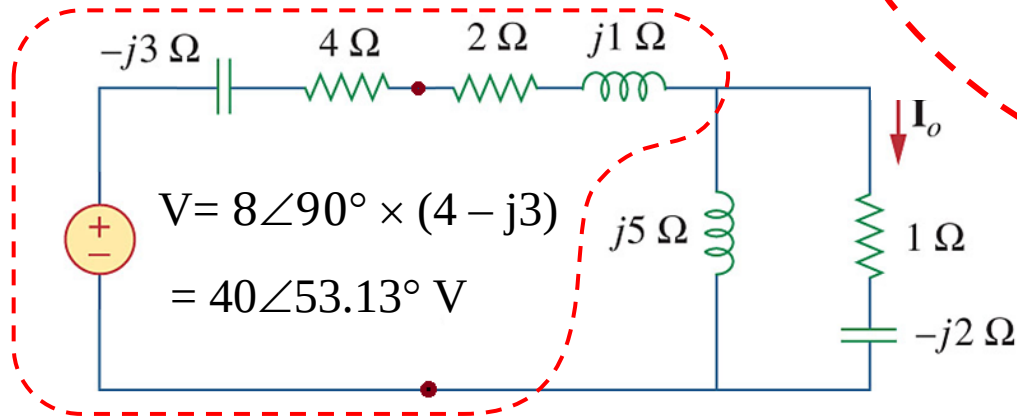


$$I = \frac{40\angle 53.13^\circ}{4 + 2 + j1 - j3} = 6.32\angle 71.57^\circ \text{ A}$$

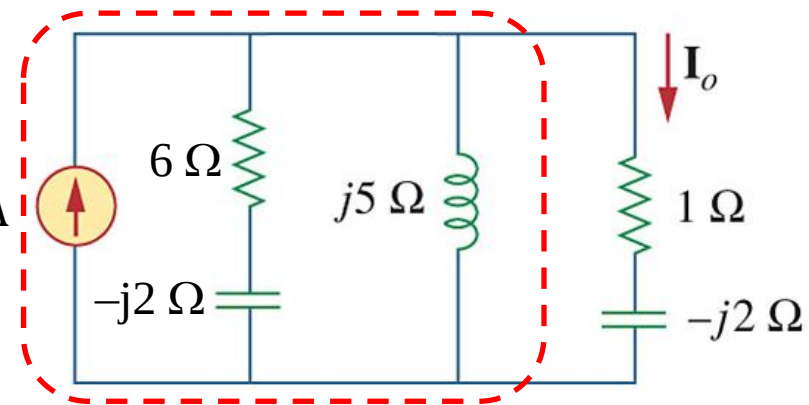
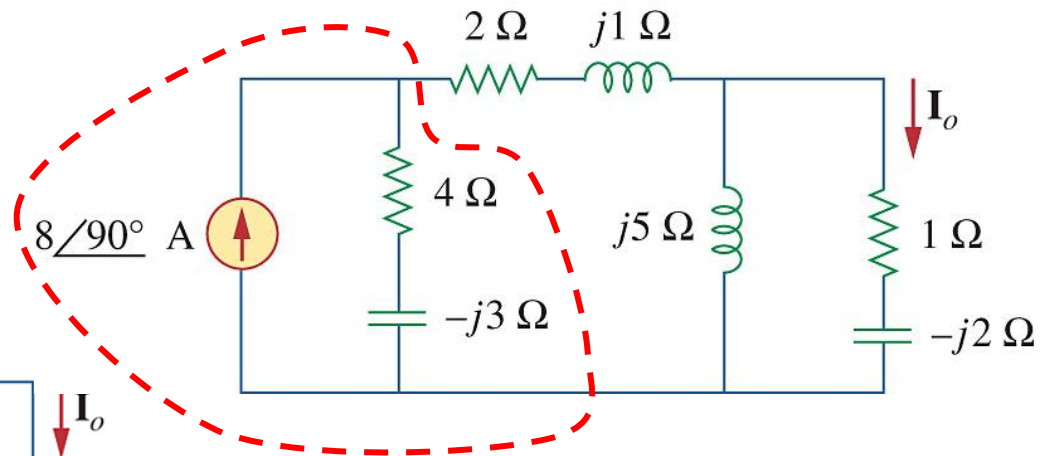


PP 10.7

Find I_o using source transformation

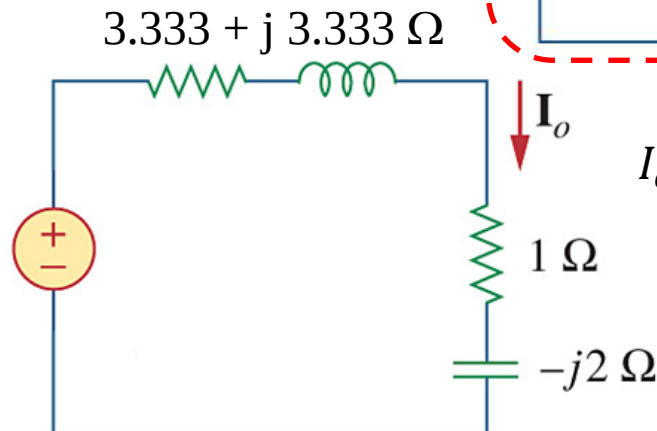


$$I = \frac{40\angle 53.13^\circ}{4 + 2 + j1 - j3} = 6.32\angle 71.57^\circ \text{ A}$$



$$V = 6.32\angle 71.57^\circ \times (6 - j2) // j5$$

$$= 29.81\angle 116.57^\circ \text{ V}$$

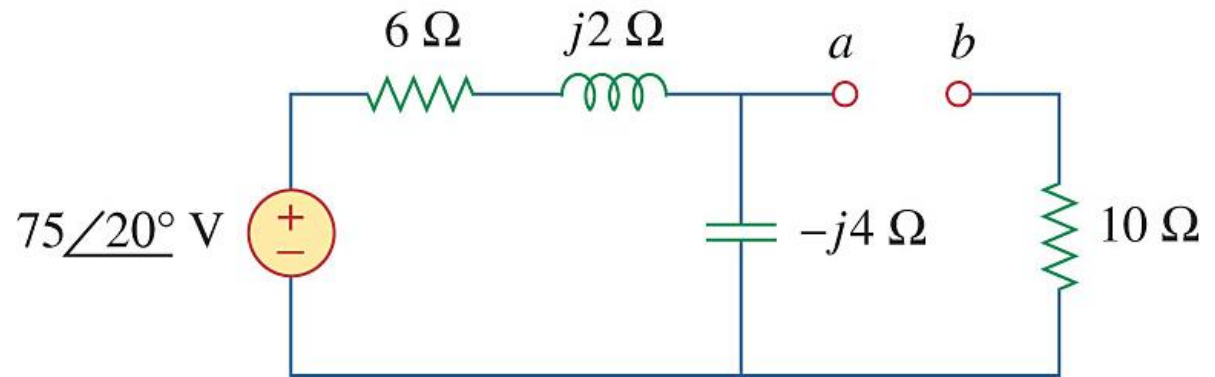


$$I_o = \frac{29.81\angle 116.57^\circ}{3.333 + 1 + j(3.333 - 2)}$$

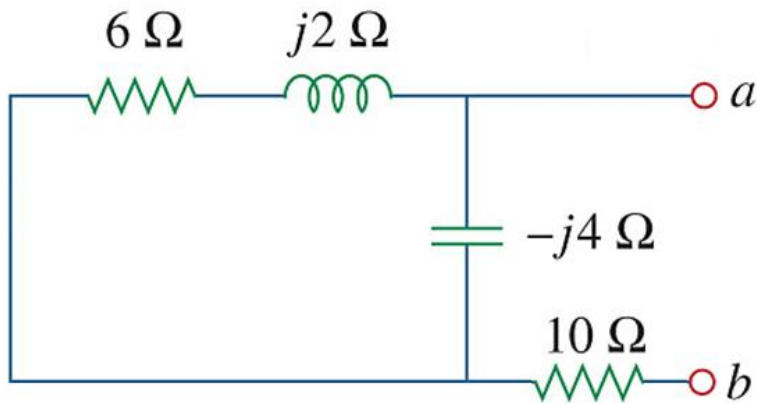
$$= \underline{6.58\angle 99.46^\circ \text{ A}}$$

PP 10.8

Find the Thevenin equivalent at terminals a-b of the circuit



Thevenin Impedance: Z_{th}



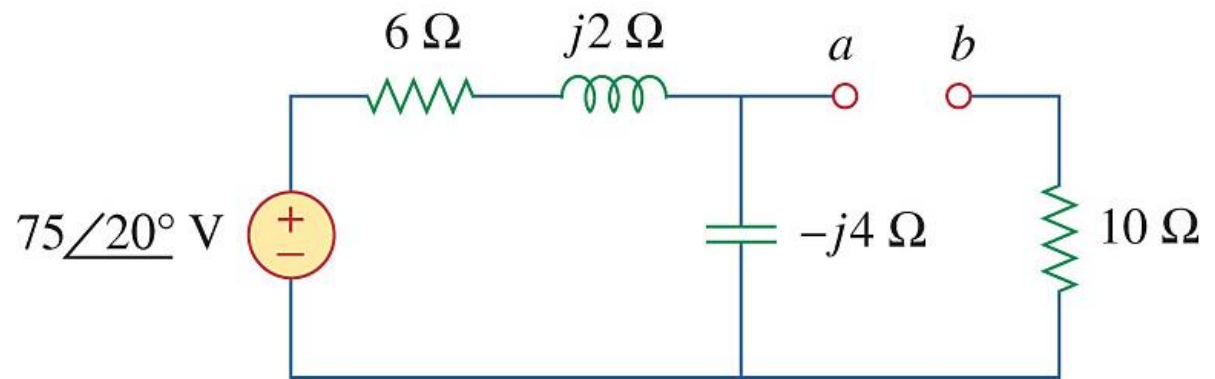
$$Z_{th} = (6 + j2) \parallel (-j4) + 10$$

$$= (12.4 - j3.2)$$

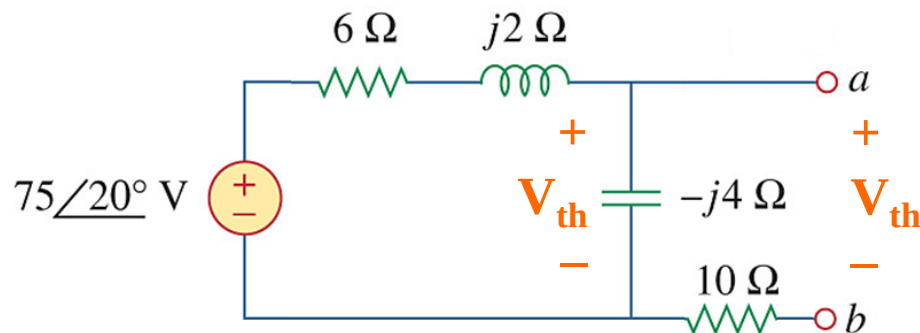
$$= \underline{12.81\angle -14.47^\circ\ \Omega}$$

PP 10.8 (cont.)

Find the Thevenin equivalent at terminals a-b of the circuit



$$V_{th} = V_{oc}$$

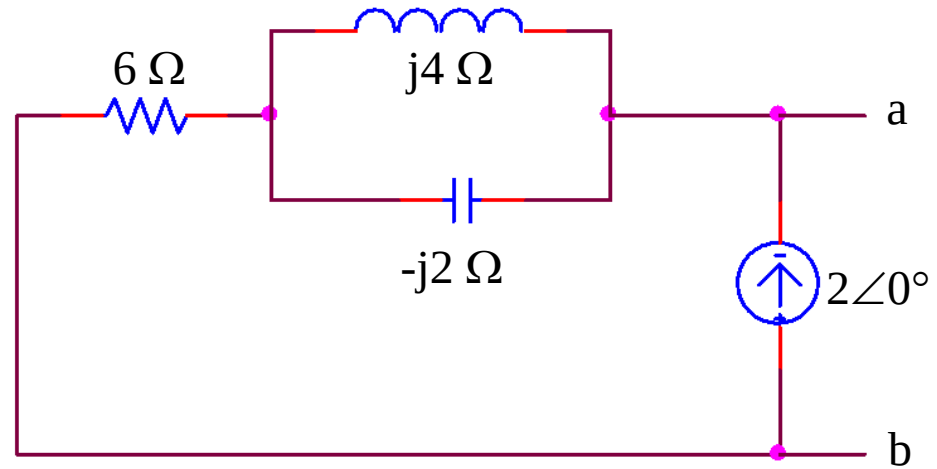


Voltage Division:

$$\begin{aligned} V_{th} &= 75\angle 20^\circ \frac{-j4}{6 + j2 - j4} \\ &= \underline{47.43\angle -51.57^\circ \text{ V}} \end{aligned}$$

Example 3

Determine the
Norton equivalent
wrt terminals a-b



Thevenin Impedance: Z_{th}
 I_{source} becomes open circuit

$$Z_{th} = j4 \parallel (-j2) + 6$$

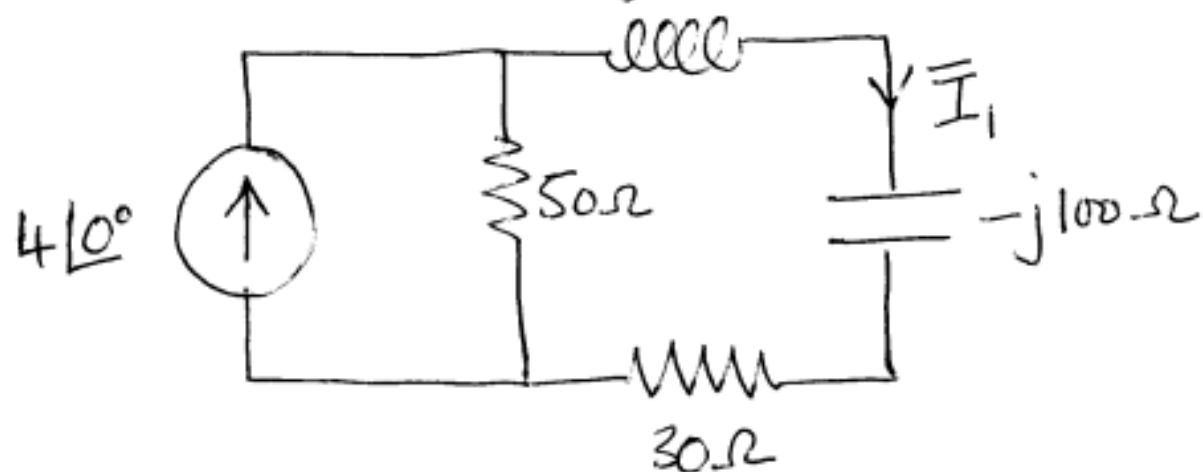
$$= (6 - j4)$$

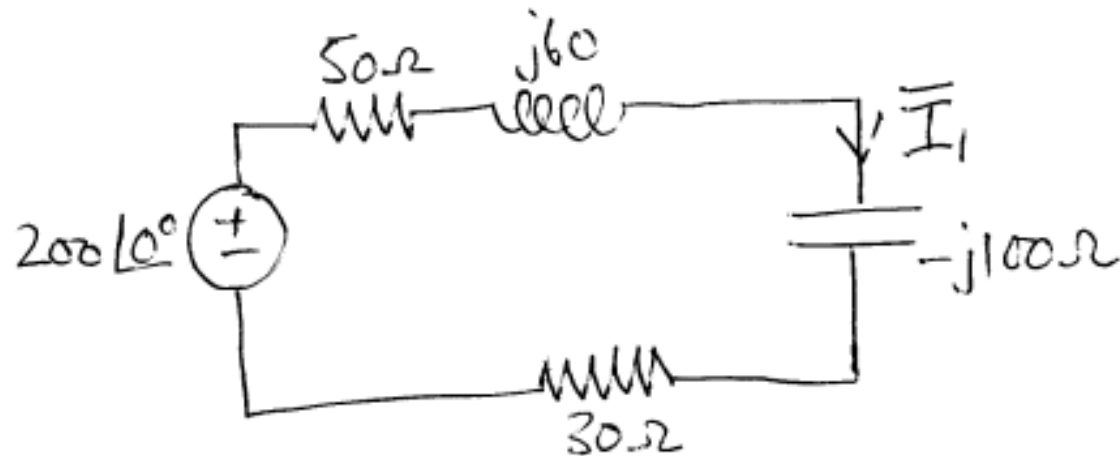
$$= 7.21\angle -33.7^\circ\ \Omega$$

$$I_N = I_{sc} = 2\angle 0^\circ$$

Question

Use source-transformation to
determine phasor $\bar{I}_1$.

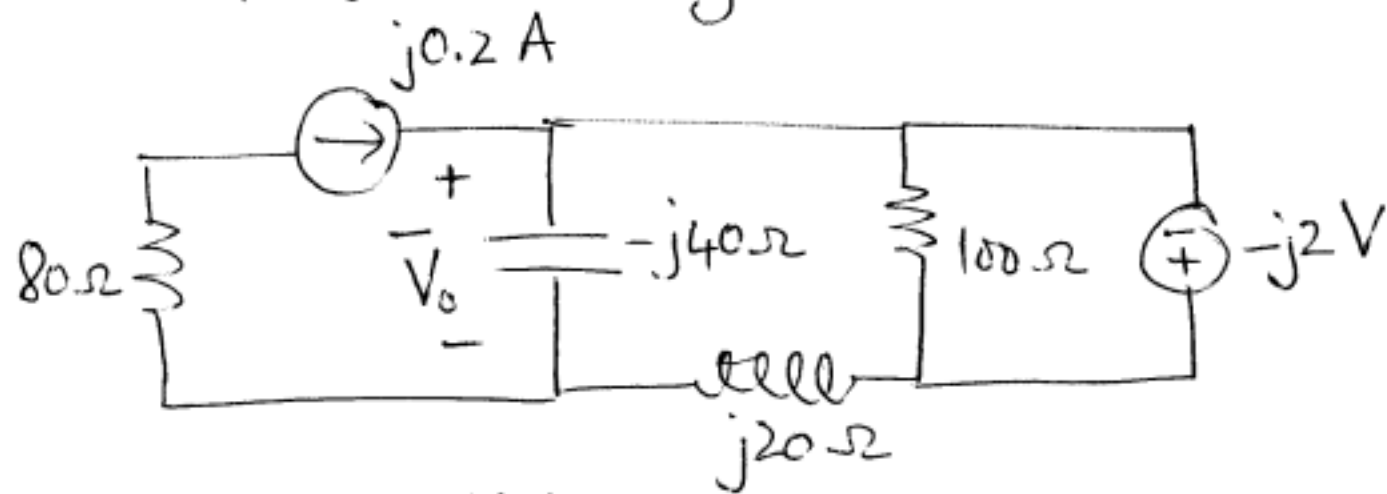


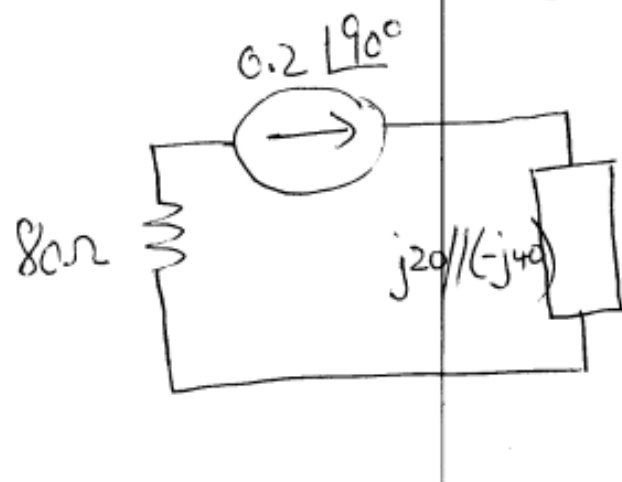
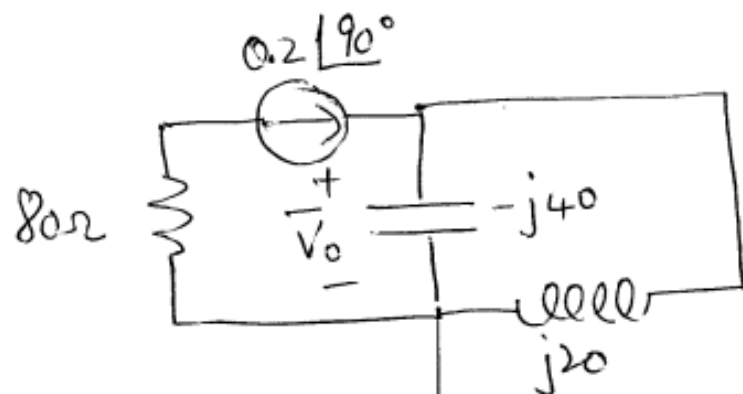
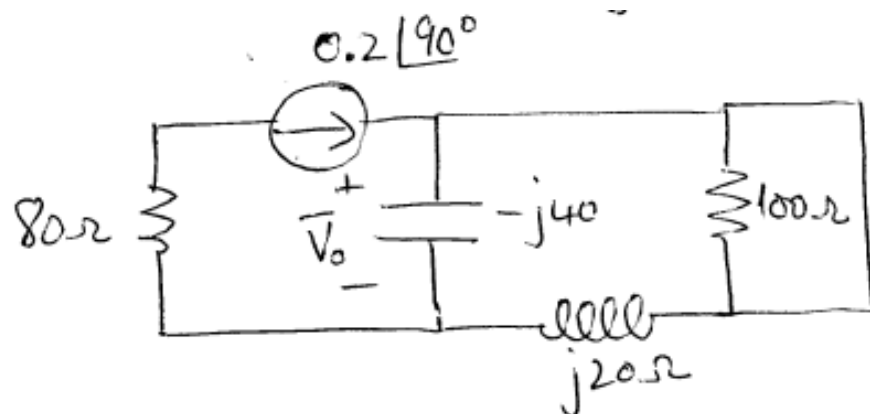


$$\begin{aligned}
 \bar{I}_1 &= \frac{200\angle 0^\circ}{80 + j60 - j100} = \frac{200\angle 0^\circ}{80 - j40} \\
 &= \frac{200\angle 0^\circ}{89.44\angle -26.6^\circ} \\
 &= 2.23\angle 26.6^\circ \text{ A}
 \end{aligned}$$

Question

Determine $\bar{V}_o$ as a result of the current source only.





$$\begin{aligned}
 & \frac{j20 \times (-j40)}{j20 - j40} \\
 &= \frac{800}{-j20} = j40 \\
 & \bar{V}_o = 0.2 \angle 90^\circ \times 40 \angle 90^\circ \\
 &= 8 \angle 180^\circ
 \end{aligned}$$

V_o